Deep Low-Rank Graph Convolutional Subspace Clustering for Hyperspectral Image

Tianhao Han, Sijie Niu, *Member, IEEE*, Xizhan Gao, Wenyue Yu, Na Cui, and Jiwen Dong

***Abstract*— Deep subspace clustering (DSC) has achieved considerable success in the classification task of hyperspectral images (HSIs) without background (defined as noisy samples) compared with traditional subspace clustering methods. Unfortunately, directly applying DSC to classify land-cover on HSI datasets with background may suffer from the degradation of classification performance. In this article, we propose an effective deep low-rank graph convolutional subspace clustering (DLR-GCSC) framework for improving the performance of land-cover classification on HSI datasets with background. Specifically, we design a joint spatial–spectral network to extract band- and patch-level features simultaneously by combining 1-D and 2-D autoencoders. Moreover, we construct a low-rank constrained fully connected layer as a self-expression layer in the network to make the joint features more discriminative. To reduce the influence of noisy samples and obtain an informative affinity matrix, we recast the joint features into a non-Euclidean domain by introducing graph convolution. Finally, spectral clustering is applied to the informative affinity matrix to obtain the classification results. Experiments on three benchmark HSI datasets show that our proposed method achieves competitive classification performance to the state-of-the-art methods on both HSI data with background and without background.**

***Index Terms*— Deep subspace clustering (DSC), graph convolutional networks (GCNs), hyperspectral image (HSI), low-rank constrained self-expression, unsupervised classification.**

I. Introduction

HYPERSPECTRAL image (HSI) is playing a more and more important role in remote sensing domain [1], [2]. Unlike RGB or multispectral images, HSI usually consists of dozens to hundreds of contiguous and narrow bands. The rich spatial and spectral information of HSI can be utilized to differentiate the region of interest (ROI) of land cover. Therefore, HSI has been successfully applied in numerous fields, such as agriculture [3], environmental monitoring [4], forest inventory [5], and military [6]. To accurately differentiate the land-cover categories, many supervised classification methods for HSI [7]–[15] have been developed over past few decades and achieved great success. However, it is generally expensive to acquire a large number of samples with high-quality annotation, which is crucial to supervised classification methods. Hence, the unsupervised learning, especially the clustering approach [16]–[19], has become ever-increasing attention for HSI.

Among all the clustering approaches, subspace clustering [20] has become a popular method for HSI classification because of the considerable performance in dealing with high-dimensional data. There are mainly two steps of subspace clustering. The first one is to learn an affinity matrix from data. Second, spectral clustering is applied to the affinity matrix to obtain the final clustering result. By minimizing certain norms of the self-representation coefficient matrix, many traditional subspace clustering methods [21]–[28] have been proposed.

The traditional subspace clustering methods have been successfully applied in unsupervised HSI classification task. For instance, in [29], a spectral–spatial sparse subspace clustering (S^4C) method integrates spectral and spatial constraint terms into a unified framework. By introducing the superpixel segmentation technique and a joint sparsity constraint, Huang *et al.* [30] designed a joint sparsity-based sparse subspace clustering (JSSC) model. Wang *et al.* [31] proposed a fast high-order sparse subspace clustering (FHoSSC) method that can capture high-order information of HSI. With the development of graph convolutional networks (GCNs) [32]–[34], Cai *et al.* [35] proposed a graph convolutional subspace clustering (GCSC) framework for HSI.

Although subspace clustering methods have achieved attractive successes, they are accompanied by the challenge that these methods are difficult to handle the data reside in a union of nonlinear subspaces [36], especially the high-dimensional HSI data. Despite a series of kernel trick works [37], [38] have been proposed to capture the nonlinear subspace structure of the data points, it is hard to ensure whether a kernel is suitable for a specific task. With the development of convolutional neural networks (CNNs), many studies have integrated traditional subspace clustering methods with CNNs. By introducing a fully connected linear layer between encoder and decoder, Ji *et al.* [36] proposed a deep subspace clustering (DSC) method to learn the affinities between all data points with a single neural network. Compared with the state-of-the-art traditional subspace clustering methods, DSC has achieved significant improvement on several standard face and object datasets. Later, many improved DSC methods have

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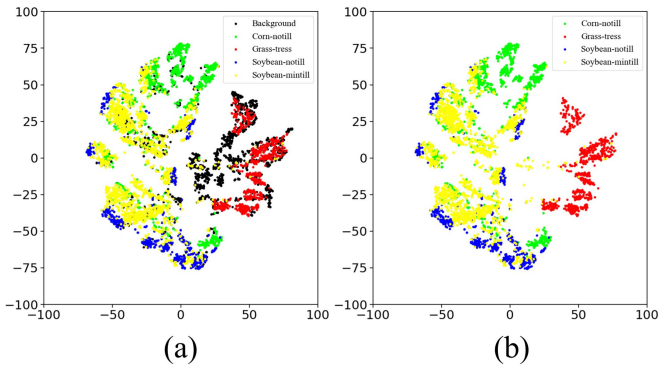


Fig. 1. Visualization of data points on the subset of the Indian Pines dataset by t-SNE [45]: (a) IPS dataset with background and (b) IPS dataset without background.

been developed to further improve the classification performance, such as Zhou *et al.* [39] used adversarial learning to learn more discriminative representations. Zhang *et al.* [40] proposed a framework to use the pseudolabel to supervise the training of the feature learning module and the self-expression module. Kheirandishfard *et al.* [41] designed a deep low-rank (DLR) architecture to learn low-rank representations by imposing a rank constraint layer.

In unsupervised HSI classification tasks, many improved DSC methods have also been developed. For example, a Laplacian regularized DSC (LRDSC) method was proposed for HSI by introducing Laplacian regularization to enhance DSC [42]. Lei *et al.* proposed a deep spatial–spectral subspace clustering network (DS^{3}C) [43] to explore the spatial–spectral information from multiscale HSI blocks. Li *et al.* proposed a self-supervised DSC with adaptive initialization method (SDSC-AI) [44] for HSI and achieved state-of-the-art unsupervised classification result. These methods have shown great potential for HSI classification tasks, but there are still some problems that need to be addressed.

In the first place, the current HSI dataset generally contains labeled regions and unlabeled regions. The data from the latter can be considered as a type of noisy sample that is generally called "background." Fig. 1 shows the t-SNE visualization on the subset of the Indian Pines dataset, indicating that the distribution of the noisy samples (background) overlaps with the labeled data points seriously. That is, the dataset contains large numbers of noisy samples, resulting in boundaries blurring between each category, which brings difficulties to the classification tasks. In the second place, the existing DSC methods for HSI classification usually discard the unlabeled information in the learning of the affinity matrix, which is improper in practice because unsupervised methods cannot obtain the labeled information and unlabeled information in advance. It has been verified in our experiments that directly applying these DSC methods to classify land-cover in HSI datasets with background may suffer from the degradation of classification performance. In the last place, the self-representation coefficient matrix learned from the self-expressive layer of DSC is directly used to construct the affinity matrix, which is difficult to characterize the relations between all data.

To solve the abovementioned problems, in this article, we propose an effective DLR-GCSC framework for improving the performance of land-cover classification on HSI datasets with background. In DLR-GCSC, we design a joint spatial–spectral network to extract band-level features and patch-level features simultaneously by combining 1-D and 2-D CAEs. Different from the self-expressive layer in existing DSC methods, we construct a low-rank constrained fully connected layer as self-expressive layer in the joint spatial–spectral network, enabling the network to learn robust joint features. Due to the presence of noisy samples (background) in HSI data, it is a challenging task for the existing DSC methods to explore an informative affinity matrix from HSI datasets with background. To alleviate the influence of noisy samples in HSI data, inspired by Cai *et al.* [35], we try to construct non-Euclidean self-representations to learn informative affinity matrix by introducing graph convolution. Finally, spectral clustering is applied to the informative affinity matrix to obtain the classification result. Importantly, the joint features are fed into GCSC to make full use of the latent graph-based global information and further improve the classification performance. Compared with state-of-the-art methods, our proposed DLR-GCSC brings significant improvements to both HSI datasets with background and without background. The main contributions can be summarized as follows.

1) We propose an end-to-end DLR-GCSC framework for unsupervised HSI classification, which can significantly improve the classification performance in both HSI datasets with background and without background.
2) A joint spatial–spectral network is designed to extract band-level features and patch-level features simultaneously and combine them into joint features. To learn more discriminative joint features, we construct a low-rank constrained fully connected layer as a self-expressive layer in the joint spatial–spectral network.
3) Different from the existing DSC methods that apply spectral-based clustering on self-representation coefficients, we introduce graph convolution to recast the joint features into the non-Euclidean domain, which alleviates the performance degradation caused by noisy samples (background).
4) Experimental results on three well-known HSI datasets demonstrate that our DLR-GCSC is better than the existing DSC methods and achieves state-of-the-art performance.

The rest of this article is organized as follows. Section II reviews the related works briefly. Section III introduces the details of our proposed DLR-GCSC method. Then, we present the experimental results and corresponding analyses in Section IV. Finally, we conclude this article in Section V.

II. Related Works

A. Sparse Subspace Clustering

Given an original data points matrix $X = [x_1, x_2, \ldots, x_N] \in \mathbb{R}^{M\times N}$, where M denotes the dimension of each data

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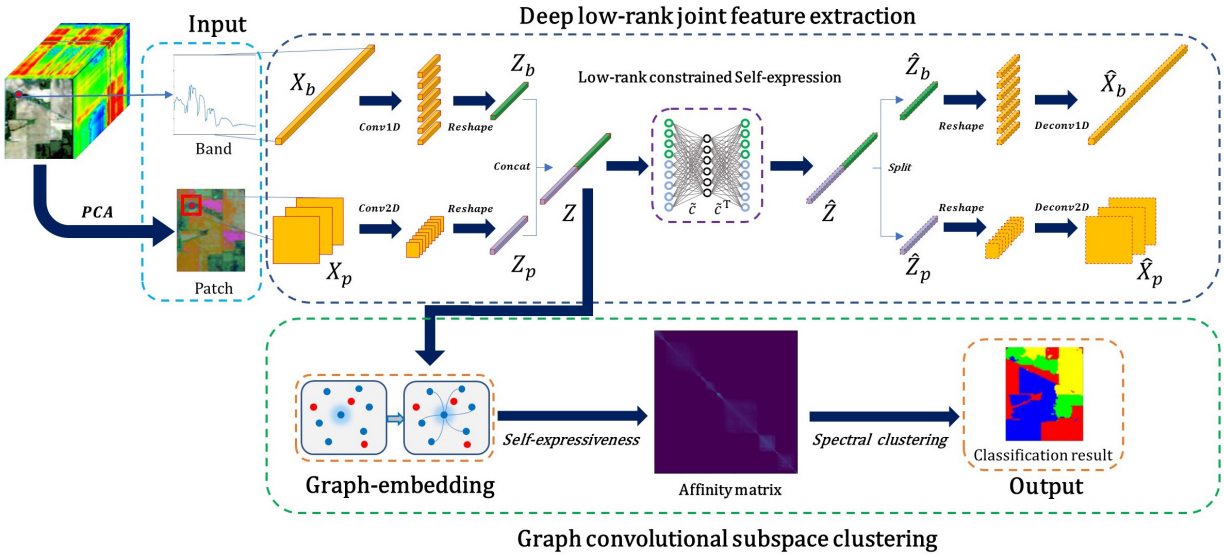


Fig. 2. Flowchart of our proposed DLR-GCSC framework.

point and N denotes the total number of data points, the data points are subject to a union of linear subspace. The self-representation can be formulated as $X = XC + E$, where $C = [c_1, c_2, \ldots, c_N] \in \mathbb{R}^{N\times N}$ denotes the self-representation coefficient matrix, and E is the representation error matrix. The constraint $\text{diag}(C) = 0$ is optionally used to avoid the trivial solution, so each sample can be represented by other sample rather than only by itself. We usually use sparse criterion [46] to obtain the matrix C by

$$\min_C \|C\|_0 \quad \text{s.t. } X = XC + E,\ \text{diag}(C) = 0 \tag{1}$$

where $\|\cdot\|_0$ is the l_0-norm that denotes the number of nonzero elements. However, due to the nonconvex property of the l_0-norm, it is usually replaced by the l_1-norm

$$\min_C \|C\|_1 \quad \text{s.t. } X = XC + E,\ \text{diag}(C) = 0. \tag{2}$$

We can solve the above problem and obtain the matrix C by using the alternating direction method of multipliers (ADMM) [47]. Then, the affinity matrix $W \in \mathbb{R}^{N\times N}$ can be constructed as

$$W = \frac{1}{2}\left(|C| + |C|^{\mathrm{T}}\right). \tag{3}$$

Finally, the classification result can be achieved by applying spectral clustering on the affinity matrix W.

B. Graph Convolutional Network

Given an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X)$, where $\mathcal{V}$ is the vertex set and $\mathcal{E}$ is the edge set, denote by $A = [a_{ij}] \in \mathbb{R}^{N\times N}$ the adjacency matrix whose elements are nonnegative. The graph Laplacian matrix L is created as $L = D - A$, and the corresponding symmetric normalized Laplacian matrix can be defined as follows:

$$\begin{aligned} L_{\text{sym}} &= D^{-\frac{1}{2}} L D^{-\frac{1}{2}} \\ &= I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \end{aligned} \tag{4}$$

where D is the diagonal matrix representing the degrees of A, and I is the identity matrix.

In the pilot work, spectral CNN [32] defines the convolution by decomposing a graph signal f on the spectral domain and then applying a spectral filter g_θ on the spectral components

$$g_\theta \star f = U g_\theta U^T f \tag{5}$$

where U is the set of eigenvectors of L and $\star$ denotes the convolution operator. To reduce the computational complexity, Defferrard *et al.* [32] proposed a K-localized ChebNet by approximating g_θ with Chebyshev polynomials up to the Kth order. The convolution can be rewritten as

$$g_\theta \star f \approx \sum_{k=0}^{K} \theta'_k T_k\left(L_{\text{sym}}\right) f \tag{6}$$

where $T_k(L_{x\text{sym}})$ is the Chebyshev polynomials and θ'_k is the Chebyshev coefficients.

Kipf and Welling [34] proposed a simplify GCN model by setting $K = 1$ and the largest eigenvalue $\lambda_{\max} = 2$; the convolution becomes

$$\begin{aligned} g_\theta \star f &\approx \theta\left(I + D^{-\frac{1}{2}} A D^{-\frac{1}{2}}\right) f \\ &= \theta\left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}\right) f \end{aligned} \tag{7}$$

where $\tilde{A} = A + I$ denotes the self-loop on A and $\tilde{D} = \sum_j \tilde{A}_{ij}$.

The propagation rule of this simple GCN model is

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right) \tag{8}$$

where $H^{(l)}$ is the output of the lth layer, $W^{(l)}$ is the trainable weight matrix in layer l, and σ is the activation function.

C. Graph Convolutional Subspace Clustering

Considering that HSI data points contain latent graph-based global information, which is neglected in the Euclidean domain, in GCSC, the HSI data points can be recast into the

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non-Euclidean domain to construct a more informative affinity matrix than traditional subspace clustering methods.

The basic idea of GCSC is to use graph convolution to modify the traditional self-representation as

$$\begin{aligned} X &= X\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\hat{C} \\ &= X\bar{A}\hat{C} \quad \text{s.t. } \operatorname{diag}(\hat{C}) = 0 \end{aligned} \tag{9}$$

where $\hat{C}$ denotes the graph convolutional self-representation coefficient matrix and $\bar{A} = \tilde{D}^{-(1/2)}\tilde{A}\tilde{D}^{-(1/2)}$ denotes the normalized adjacency matrix of the embedded graph. The GCSC aims to solve the problem as follows:

$$\min_{\hat{C}} \frac{1}{2}\left(\lambda\|\hat{C}\|_p + \|X - X\bar{A}\hat{C}\|_q\right) \quad \text{s.t. } \operatorname{diag}(\hat{C}) = 0 \tag{10}$$

where λ is the tradeoff coefficient, and $\|\cdot\|_p$ and $\|\cdot\|_q$ represent any appropriate matrix norm, such as $p, q = 0, 1, 2$ and $*$. Finally, the classification results can be obtained through spectral clustering algorithm.

D. Deep Subspace Clustering

Let Θ represent the parameters of the autoencoder. Specifically, Θ consists of the encoder parameters Θ_e and the decoder parameters Θ_d. The loss function of DSC can be defined as

$$\begin{aligned} L(\Theta, C) = &\frac{1}{2}\left(\|X - \hat{X}_\Theta\|_F^2 + \lambda\|Z_{\Theta_e} - Z_{\Theta_e}C\|_F^2\right) \\ &+\gamma\|C\|_p \quad \text{s.t. } \operatorname{diag}(C) = 0 \end{aligned} \tag{11}$$

where $\hat{X}_\Theta$ denotes the data reconstructed by the autoencoder and Z_{Θ_e} is the output of the encoder, which denotes the latent feature mapped from the data points matrix X. Now, the self-representation coefficient matrix C can be directly learned by a neural network.

III. Proposed Method

The flowchart of our proposed DLR-GCSC is illustrated in Fig. 2; DLR-GCSC mainly consists of two parts: DLR features' extraction and joint feature-based GCSC. We will introduce the details of each part and then illustrate the implementation details in this section.

A. Deep Low-Rank Joint Features' Extraction

1) Joint Spatial–Spectral Network: Because the neighboring pixels in HSI have a high possibility of belonging to the same category, the existing DSC methods usually use 2-D CAE to extract the features from patches. However, the contribution of spectral information at the center of the patches may be reduced, which leads to a lack of robustness when faced with noisy samples. Therefore, we add a 1-D CAE to the network to make full use of the spectral information of HSI. The spectral features extracted from bands by 1-D CAE and the spatial–spectral features extracted from patches by 2-D CAE are concatenated to build more robust joint spatial–spectral features. Figs. 3 and 4 show the network architecture of the encoder and the decoder, respectively. The detailed settings are shown in Table I.

Let a matrix $X_b = [x_{b_1}, x_{b_2}, \ldots, x_{b_n}] \in \mathbb{R}^{d\times n}$ be a collection of band samples $\{x_{b_i} \in \mathbb{R}^d\}_{i=1}^n$, where d and n denote the dimension and the number of samples, respectively. Considering that the dimension of HSI data is very high, we first apply the principal component analysis (PCA) to reduce the dimension of X_b to $\hat{d}$. Then, a rolling window operation is used to construct $w \times w$ spatial neighborhood patches for each pixel, where w is the size of patch. We can obtain the spatial neighborhood patches $X_p = [X_{p_1}, X_{p_2}, \ldots, X_{p_n}] \in \mathbb{R}^{w\times w\times \hat{d}\times n}$.

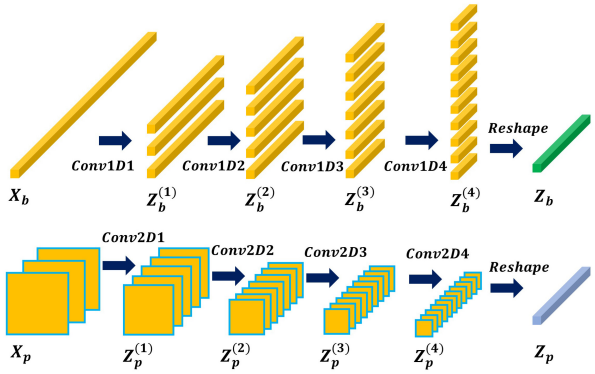


Fig. 3. Network architecture of the encoder.

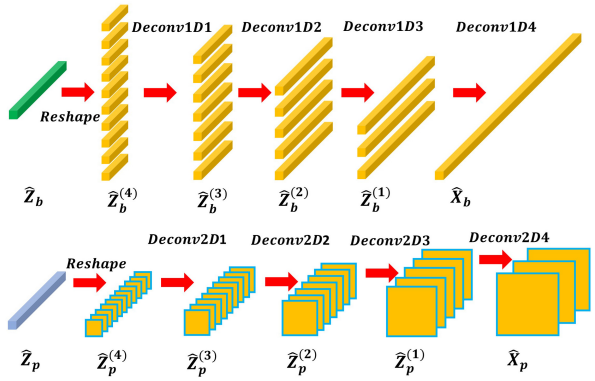


Fig. 4. Network architecture of the decoder.

In our proposed method, X_b is mapped into the latent feature matrix $Z_b = [z_{b_1}, z_{b_2}, \ldots, z_{b_n}] \in \mathbb{R}^{l_b\times n}$ through 1-D encoder and X_p is mapped into $Z_p = [Z_{p_1}, Z_{p_2}, \ldots, Z_{p_n}] \in \mathbb{R}^{w_l\times w_l\times d_l\times n}$ through the 2-D encoder, respectively, where l_b denotes the length of z_{b_i}, w_l denotes the scale of z_{p_i}, and d_l denotes the latent channel of z_{p_i}.

Let Θ_{e_b} denote the 1-D encoder network parameters and $f_{\Theta_{e_b}}(\cdot)$ denote the map function of the 1-D encoder. We can formulate the operation of the 1-D encoder as

$$Z_b^{(4)} = f_{\Theta_{e_b}}(X_b). \tag{12}$$

Let Θ_{e_p} denote the 2-D encoder network parameters and $f_{\Theta_{e_p}}(\cdot)$ denote the map function of the 2-D encoder. We can formulate the operation of the 2-D encoder as

$$Z_p^{(4)} = f_{\Theta_{e_p}}(X_p). \tag{13}$$

Since $Z_{p_i}^{(4)} \in \mathbb{R}^{w_l\times w_l\times d_l}$ is a 3-D cube, which denotes the feature map of latent layer, we need to vectorize the $Z_{p_i}^{(4)}$ into a column vector $z_{p_i} \in \mathbb{R}^{l_p}$ and $Z_p = [z_{p_1}, z_{p_2}, \ldots, z_{p_n}] \in \mathbb{R}^{l_p\times n}$, where $l_p = w_l \times w_l \times d_l$.

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TABLE I
NETWORK SETTINGS OF DLR-GCSC

1D CAE				2D CAE			
Layers	Kernel	Channels	Stride	Layers	Kernel	Channels	Stride
Conv1D1	5	16	3	Conv2D1	5×5	16	2×2
Conv1D2	5	16	2	Conv2D2	5×5	32	2×2
Conv1D3	5	32	2	Conv2D3	5×5	64	2×2
Conv1D4	5	32	2	Conv2D4	5×5	128	2×2
Deconv1D1	5	32	2	Deconv2D1	5×5	128	2×2
Deconv1D2	5	32	2	Deconv2D2	5×5	64	2×2
Deconv1D3	5	16	2	Deconv2D3	5×5	32	2×2
Deconv1D4	5	16	3	Deconv2D4	5×5	16	2×2

We vertically concatenate Z_b with Z_p and obtain a joint spatial–spectral feature matrix

$$Z = \left[\left(z_{b_1} : z_{p_1}\right), \left(z_{b_2} : z_{p_2}\right), \ldots, \left(z_{b_n} : z_{p_n}\right)\right] \in \mathbb{R}^{l\times n} \quad (14)$$

where (:) denotes the vertically concatenate operation and $l = l_b + l_p$ denotes the length of the joint spatial–spectral feature vector.

Then, the self-expressive feature matrix $\hat{Z}$ is obtained by the self-expressive layer. Based on the reconstruction criterion, $\hat{Z}$ can be formulated as

$$\hat{Z} = \left[\left(\hat{z}_{b_1} : \hat{z}_{p_1}\right), \left(\hat{z}_{b_2} : \hat{z}_{p_2}\right), \ldots, \left(\hat{z}_{b_n} : \hat{z}_{p_n}\right)\right] \in \mathbb{R}^{l\times n} \quad (15)$$

where the shapes of $\hat{z}_{b_i}$ and $\hat{z}_{p_i}$ are the same as z_{b_i} and z_{p_i}, respectively, so we can separate $\hat{Z}_b$ and $\hat{Z}_p$ from $\hat{Z}$. Let Θ_{d_b} denotes the 1-D decoder network parameters, Θ_{d_p} denote the 2-D decoder network parameters, $f_{\Theta_{d_b}}(\cdot)$ denote the deconvolution operation of the 1-D decoder, and $f_{\Theta_{d_p}}(\cdot)$ denote the deconvolution operation of the 2-D decoder. We can get the reconstruction by

$$\hat{X}_b = f_{\Theta_{d_b}}\left(\hat{Z}_b^{(4)}\right) \quad (16)$$

$$\hat{X}_p = f_{\Theta_{d_p}}\left(\hat{Z}_p^{(4)}\right) \quad (17)$$

where $\hat{X}_b \in \mathbb{R}^{m\times n}$ denotes the reconstructed bands and $\hat{X}_p \in \mathbb{R}^{w\times w\times d\times n}$ denotes the reconstructed patches. Therefore, the loss function for this features' extraction process is the sum of two reconstruction errors

$$\mathcal{L}_{\text{rec}} = \frac{1}{2}\|X_b - \hat{X}_b\|_F^2 + \frac{1}{2}\|X_p - \hat{X}_p\|_F^2. \quad (18)$$

2) Low-Rank Constrained Self-Expressive Layer: To learn more discriminative latent features than the original DSC network, we utilize low-rank regularization to constrain the matrix C. Usually, the low-rank constrained self-expressiveness problem can be formulated as

$$\min_C \lambda\|C\|_* + \gamma\,\|Z - ZC\|_F^2 \quad (19)$$

where $\|C\|_*$ denotes the nuclear norm on self-representation coefficient matrix. Since the LRR utilized singular value decomposition (SVD) to solve the nuclear norm $\|C\|_*$ in the objective function, it is hard to combine LRR with deep learning because of the high computational cost. Inspired by DLRSC [41], we replace the self-expressive layer in the DSC module with two fully connected layer whose parameters are $\tilde{c}$ and its transpose $\tilde{c}^{\mathrm{T}}$, respectively. Then, we can obtain a symmetric self-representation coefficient matrix C by $C = \tilde{c}\tilde{c}^{\mathrm{T}}$, where $\tilde{c} \in \mathbb{R}^{n\times r}$ is the weight matrix and r ($r \ll n$) is used to limit the maximum possible rank of the matrix C. We can conclude that $rank(C) \le r$ is always true, which is equivalent to pose a low-rank constraint on C. The loss function of this low-rank constrained self-expressive layer L_{se} can be formulated as

$$\mathcal{L}_{se} = \lambda\|\tilde{c}\tilde{c}^{\mathrm{T}}\|_2 + \gamma\,\|Z - Z\tilde{c}\tilde{c}^{\mathrm{T}}\|_F^2 \quad (20)$$

where λ and γ are the tradeoff parameters to balance the self-expression loss.

The total loss function of the network is defined as

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{rec} + \mathcal{L}_{se} \\ &= \frac{1}{2}\left(\|X_b - \hat{X}_b\|_F^2 + \|X_p - \hat{X}_p\|_F^2\right) \\ &\quad + \lambda\|\tilde{c}\tilde{c}^{\mathrm{T}}\|_2 + \gamma\,\|Z - Z\tilde{c}\tilde{c}^{\mathrm{T}}\|_F^2. \end{aligned} \quad (21)$$

Note that the elements of $\tilde{c}$ that is learned in DLR-GCSC ($n \times r$) are much fewer than the elements of C that is learned in DSC-based methods (n^2), which results in a cheaper computational cost in DLR-GCSC.

B. Joint Feature-Based Graph Convolutional Subspace Clustering

In conventional DSC methods, the learned affinity matrix is vulnerable to noisy samples when directly applied to the self-representation coefficients to construct the affinity matrix, resulting in classification performance degradation. Inspired by Cai *et al.* [35], we employ GCSC to recast the joint spatial–spectral feature matrix Z into a non-Euclidean domain to exploit the graph-based global information and constrict an informative affinity matrix, which is a benefit to further improve the classification performance on HSI datasets with background.

We first obtain Z from the DLR joint features' extraction process and then construct a k-nearest neighbor (KNN) graph to describe the graph-based global similarity of the data points. We can obtain the adjacent matrix A of the KNN graph $\mathcal{G}$; each element of A can be defined as

$$A_{ij} = \begin{cases} 1, & z_j \in \mathcal{N}_k(z_i) \\ 0, & z_j \notin \mathcal{N}_k(z_i) \end{cases} \quad (22)$$

where z_i and z_j are the column of Z, and $\mathcal{N}_k(z_i)$ denotes a set that contains k neighbors of z_i. If $A_{ij} = 1$, z_i and z_j are similar, so the ith sample and the jth sample are likely lying in the same subspace.

After obtaining the adjacent matrix, the graph convolution feature matrix E is obtained by

$$\begin{aligned} E &= Z\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\right) \\ &= Z\bar{A} \end{aligned} \quad (23)$$

where $\tilde{A} = A + I$ is to add self-connections on A, $\tilde{D}$ is a degree matrix of $\tilde{A}$, and $\bar{A} = \tilde{D}^{-(1/2)}\tilde{A}\tilde{D}^{-(1/2)}$ denotes the normalized adjacency with self-loops.

To learn an informative affinity, the traditional self-representation can be modified as follows:

$$\begin{aligned} Z &= Z\bar{A}\hat{C} \\ &= E\hat{C} \quad \text{s.t. } \operatorname{diag}\left(\hat{C}\right) = 0 \end{aligned} \quad (24)$$

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where $\hat{C}$ denotes the graph convolutional self-representation coefficient matrix; the objective function of GCSC can be rewritten as

$$\min_{\hat{C}} \frac{1}{2}\left(\lambda_g \|\hat{C}\|_p + \|Z - E\hat{C}\|_q\right) \quad \text{s.t. } \operatorname{diag}(\hat{C}) = 0. \tag{25}$$

By setting $\|\cdot\|_p$ and $\|\cdot\|_q$ to be the Frobenius norm, we can get the objective function as follows:

$$\begin{aligned}\mathcal{L}(\hat{C}) &= \frac{\lambda_g}{2}\|\hat{C}\|_F^2 + \frac{1}{2}\|Z - E\hat{C}\|_F^2 \\ &= \frac{1}{2} tr\left[\lambda_g \hat{C}^{\mathrm{T}}\hat{C} + \left(Z - E\hat{C}\right)^{\mathrm{T}}\left(Z - E\hat{C}\right)\right] \\ &= \frac{1}{2} tr\left(\lambda_g \hat{C}^{\mathrm{T}}\hat{C} + Z^{\mathrm{T}}Z + \hat{C}\hat{C}^{\mathrm{T}}E^{\mathrm{T}}E\hat{C} - 2Z^{\mathrm{T}}E\hat{C}\right).\end{aligned} \tag{26}$$

The partial derivative of $\mathcal{L}(\hat{C})$ with respect to $\hat{C}$ is

$$\frac{\partial \mathcal{L}(\hat{C})}{\partial \hat{C}} = \left(\lambda_g I + E^{\mathrm{T}}E\right)\hat{C} - E^{\mathrm{T}}Z. \tag{27}$$

Algorithm 1 Our Proposed DLR-GCSC Method for HSI Classification

1: **Input:** Bands X_b and patches X_p created from the normalized HSI data X, tradeoff parameters λ, γ, and λ_g, the number of nearest neigubors k for graph embedding, the rank scalar r, and the maximum training epochs T_{max}
2: **Begin**
3: Initialize the weight and bias of the network.
4: **while** $epoch \leq T_{max}$ **do**
5: Feed X_b into 1D band auto-encoder and X_p into 2D patch auto-encoder to get Z_b and Z_p.
6: Concatenate Z_b with Z_p to get the joint spatial-spectral feature Z, update Z.
7: Z is mapped to $\hat{Z}$ after the low-rank constrained self-expressive layer.
8: Split $\hat{Z}$ into $\hat{Z}_b$ and $\hat{Z}_p$ as the input shape, through 1D decoder and 2D decoder to obtain the reconstruction $\hat{X}_b$ and $\hat{X}_p$.
9: Update parameters of the network by minimizing Eq. (21).
10: **end while**
11: Perform graph convolutional subspace clustering process to get the classification result Y.
12: **End**
13: **Output:** HSI classification result Y corresponding to X.

Let $(\partial \mathcal{L}(\hat{C})/\partial \hat{C}) = 0$; we can get $\hat{C}$ by

$$\hat{C} = \left(\lambda_g I + E^{\mathrm{T}}E\right)^{-1} E^{\mathrm{T}}Z \tag{28}$$

where $\lambda_g I + E^{\mathrm{T}}E$ is positive semidefinite, so the reversible of it always exists.

The obtained graph convolutional self-representation coefficient matrix $\hat{C}$ can be applied in constructing the affinity matrix S as follows:

$$S = \frac{1}{2}\left(|\hat{C}| + |\hat{C}|^{\mathrm{T}}\right). \tag{29}$$

Then, we apply spectral clustering to the affinity matrix S to get the final classification result Y. Notice that spectral clustering is an unsupervised method; the classification label index assigned to each cluster might not be consistent with the ground truth. Thus, we perform a permutation of Y by the Hungarian algorithm [48].

TABLE II
NUMBER OF EACH CLASS FOR IPS, PUS, AND SA DATASETS

IPS		PUS		SA	
Class	Sample	Class	Sample	Class	Sample
Corn-notill	1005	Asphalt	425	Brocoli green weeds_1	391
Grass-tree	730	Meadows	768	Corn senesced green	1343
Soybean-notill	732	Trees	63	Lettuce romaine_4wk	616
Soybean-mintill	1924	Metal sheet	1315	Lettuce romaine_5wk	1525
Unlabeled	**1559**	Bare soil	2559	Lettuce romaine_6wk	674
		Bitumen	860	Lettuce romaine_7wk	799
		Bricks	94	**Unlabeled**	**1790**
		Shadows	361		
		Unlabeled	**13555**		

C. Implementation Details

We first preprocess the HSI data with dimensionality reduction, patch construction, and data normalization. The preprocessed HSI data are then fed into our proposed framework. The network of our proposed method is optimized by adaptive moment estimation (Adam) [49] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and the learning rate is set to 0.0001. The detailed training procedure of our DLR-GCSC is illustrated in Algorithm 1.

IV. EXPERIMENTS

In this section, we assess the classification performance of our proposed method on three widely used HSI datasets. The experiments are implemented in Python 3.7 and run on a single TITAN RTX GPU.

A. Experimental Setting

1) Datasets' Description: We conduct our experiments on three real HSI datasets (i.e., the Indian Pines dataset, the Pavia University dataset, and the SalinasA (SA) dataset). Since the subspace clustering method is difficult to deal with large-scale datasets, both traditional subspace clustering methods and DSC methods are usually conducted on small-size HSIs. Thus, we adopt the subset of Indian Pines and Pavia University as [35] and [44] do, and the subsets are located within the original datasets at [30–115, 24–94] and [150–350, 100–200], respectively.

a) Indian pines subset (IPS): The IPS dataset was obtained by the Airborne Visible Imaging Spectrometer (AVIRIS) sensor. It consists of 5950 pixels and 200 spectral bands (the water absorption bands 104–108, 150–163, and 220 are removed) in the wavelength range from 0.2 to 2.4μm. Four labeled classes are corn-notill, grass-trees, soybean-notill, and soybean-mintill, respectively. For the hyperparameters of the IPS dataset, we set tradeoff parameters $\lambda = 1$, $\gamma = 100$, $\lambda_g = 10$, rank scalar $r = 320$, nearest neighbors $k = 60$, patch window size $w = 27$, and principal components $d = 3$.

b) Pavia university subset (PUS): The PUS dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. It consists of 20 000 pixels and 103 spectral bands (12 noisy bands are removed) with a spatial resolution of 1.3 m per pixel. Eight labeled classes are asphalt, meadows, trees, metal sheet, bare soil, bitumen, bricks, and shadows, respectively. For the hyperparameters of the PUS dataset, we set $\lambda = 0.001$, $\gamma = 100$, $\lambda_g = 10$, $r = 400$, $k = 160$, $w = 31$, and $d = 4$.

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TABLE III
CLASSIFICATION RESULTS OF EKGCSC, SSC, DSC, DS^{3}C, S^{2}CONVSCN, SDSC-AI, JSSC, SSC+, FHOSSC, DGCSC, AND DLR-GCSC ON THE IPS DATASET

Class	EKGCSC	SSC	DSC	DS^{3}C	S^{2}ConvSCN	SDSC-AI	JSSC	SSC+	FHoSSC	DGCSC	DLR-GCSC
Corn-notill(%)	0.00	62.49	64.38	49.15	62.39	88.56	74.23	70.25	70.15	90.75	74.63
Grass-trees(%)	80.27	99.59	100	99.86	85.89	95.34	99.86	100	100	71.37	100
Soybean-notill(%)	99.59	70.63	76.09	66.53	81.15	81.83	85.93	100	100	99.86	100
Soybean-mintill(%)	75.88	57.33	76.77	83.63	87.27	79.94	88.25	87.16	91.73	94.13	95.43
$OA(\%)$	63.20	67.75	77.68	75.59	80.32	84.79	86.59	87.57	89.39	90.53	**92.19**
$AA(\%)$	63.94	72.51	79.31	74.79	79.18	86.42	87.07	89.35	90.47	89.03	**92.52**
$Kappa(\%)$	51.57	55.78	69.21	64.14	71.82	78.71	80.91	82.52	84.97	86.44	**88.90**

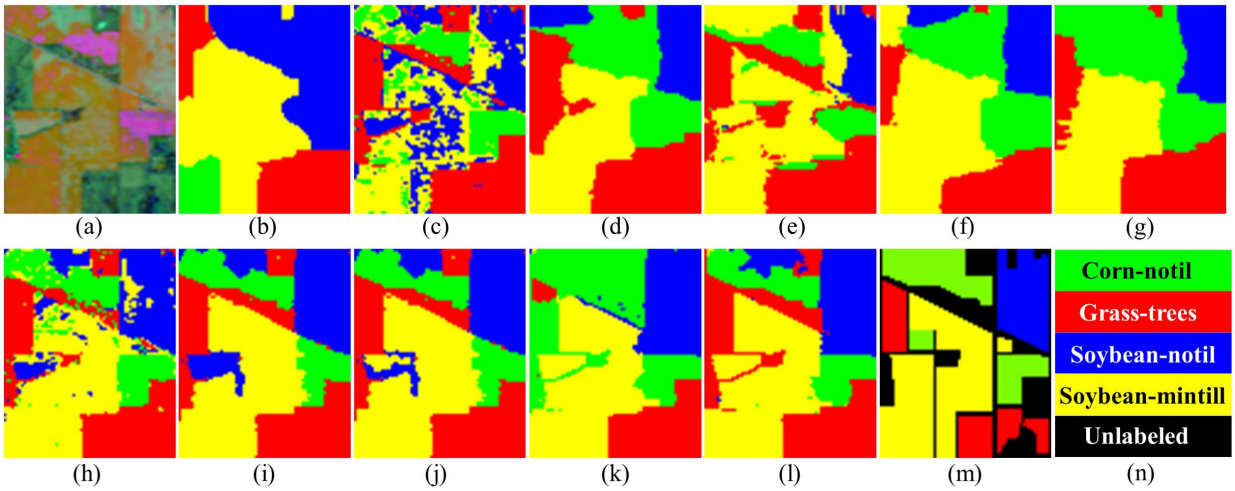


Fig. 5. Classification maps of the different methods with the IPS dataset (with background): (a) false color image, (b) EKGCSC, (c) SSC, (d) DSC, (e) DS^{3}C, (f) S^{2}ConvSCN, (g) SDSC-AI, (h) JSSC, (i) SSC+, (j) FHoSSC, (k) DS-GCSC, (l) our proposed DLR-GCSC, (m) ground truth, and (n) label.

c) SalinasA: The SA dataset was acquired by the AVIRIS sensor. It consists of 7138 pixels and 204 spectral bands (the water absorption bands 108–112, 154–167, and 224 are removed) with a spatial resolution of 3.7m per pixel. Six labeled classes are brocoli green weeds_1, corn senesced green, lettuce romaine_4wk, lettuce romaine_5wk, lettuce romaine_6wk, and lettuce romaine_7wk, respectively. For the hyperparameters of the SA dataset, we set $\lambda = 0.001$, $\gamma = 100$, $\lambda_g = 10$, $r = 360$, $k = 120$, $w = 27$, and $d = 5$.

The detailed categories are shown in Table II. It is noteworthy that the unlabeled samples are not regarded as a new category but an essential part of an HSI dataset with background.

2) Compared Methods: We compare our DLR-GCSC with several traditional subspace clustering methods and DSC methods to assess the classification performance. Traditional subspace clustering methods include SSC [46], JSSC [30], SSC+ [31], FHoSSC [31], and EKGCSC [35]. DSC methods include DSC [36], S^{2}ConvSCN [40], DS^{3}C [43], and SDSC-AI [44]. In addition, a variant of our proposed method called deep GCSC (DGCSC) is also performed to compare the classification performance. DGCSC uses the original DSC self-expressive layer to prove the effectiveness of our DLR-GCSC method. We utilize EKGCSC, SSC, DSC, DS^{3}C, S^{2}ConvSCN, SDSC-AI, JSSC, SSC+, FHoSSC, and DGCSC to conduct the experiments on HSI datasets with background (i.e., the input contains unlabeled data) and use RMMF, DSC, DS^{3}C, S^{2}ConvSCN, EKGCSC, SDSC-AI, and DGCSC to conduct the experiments on HSI datasets without background (i.e., the input does not contain unlabeled data). The parameters of all compared methods are followed by their settings, which are reported in their corresponding literature.

3) Metrics: We also employ three commonly used metrics to evaluate the classification performance, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa), given a confusion matrix $T \in \mathbb{R}^{l\times l}$, where l denotes the number of categories.

OA can be defined as

$$\mathrm{OA} = \frac{\sum_{i=1}^{l} T_{ii}}{P} \tag{30}$$

where P denotes the total number of the samples and T_{ii} is the Diagonal element of T.

AA can be defined as

$$\mathrm{AA} = \frac{\sum_{j=1}^{l} \left(T_{jj} / \sum_{j=1}^{l} T_{ij}\right)}{l} \tag{31}$$

where T_{ij} denotes the number of samples, which should be classified as class i but be classified as class j.

Kappa can be defined as

$$\mathrm{Kappa} = \frac{P\sum_{i=1}^{l} T_{ii} - \sum_{i=1}^{l}\left(\sum_{j=1}^{l} T_{ij} \sum_{j=1}^{l} T_{ji}\right)}{P^2 - \sum_{i=1}^{l}\left(\sum_{j=1}^{l} T_{ij} \sum_{j=1}^{l} T_{ji}\right)}. \tag{32}$$

B. Results and Analysis on HSI Datasets With Background

In the first experiment, we evaluate the performance of our proposed DLR-GCSC on three HSI datasets with background. Tables III–V report the quantitative classification results and

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TABLE IV
CLASSIFICATION RESULTS OF EKGCSC, SSC, DSC, DS^{3}C, S^{2}CONVSCN, SDSC-AI, JSSC, SSC+, FHOSSC, DGCSC, AND DLR-GCSC ON THE PUS DATASET

Class	EKGCSC	SSC	DSC	DS^{3}C	S^{2}ConvSCN	SDSC-AI	JSSC	SSC+	FHoSSC	DGCSC	DLR-GCSC
Asphalt(%)	0.00	6.12	68.71	13.18	90.59	6.59	11.76	0.47	0.00	0.00	100
Meadows(%)	100	57.55	0.00	100	0.00	100	99.35	99.09	99.87	100	100
Trees(%)	0.00	95.25	100	39.68	100	63.49	98.41	92.06	93.65	79.37	80.95
Metal Sheet(%)	100	98.10	100	99.47	100	99.70	99.77	97.34	96.73	99.62	99.39
Bare soil(%)	96.64	39.12	91.91	71.90	90.04	91.87	68.50	81.01	81.56	96.37	100
Bitumen(%)	100	98.49	100	79.65	100	100	99.07	100	100	100	100
Bricks(%)	0.00	0.00	8.51	0.00	11.70	0.00	0.00	0.00	0.00	0.00	0.00
Shadows(%)	0.00	82.55	0.00	45.15	0.00	59.28	89.02	94.18	100	100	59.28
$OA(\%)$	84.03	61.51	75.87	75.17	76.62	86.45	79.35	83.38	83.88	90.22	**95.95**
$AA(\%)$	49.58	59.65	58.64	56.13	61.54	65.12	70.74	70.52	71.48	71.92	**79.95**
$Kappa(\%)$	78.67	53.78	66.97	68.60	68.31	82.30	74.04	78.70	79.33	87.14	**94.65**

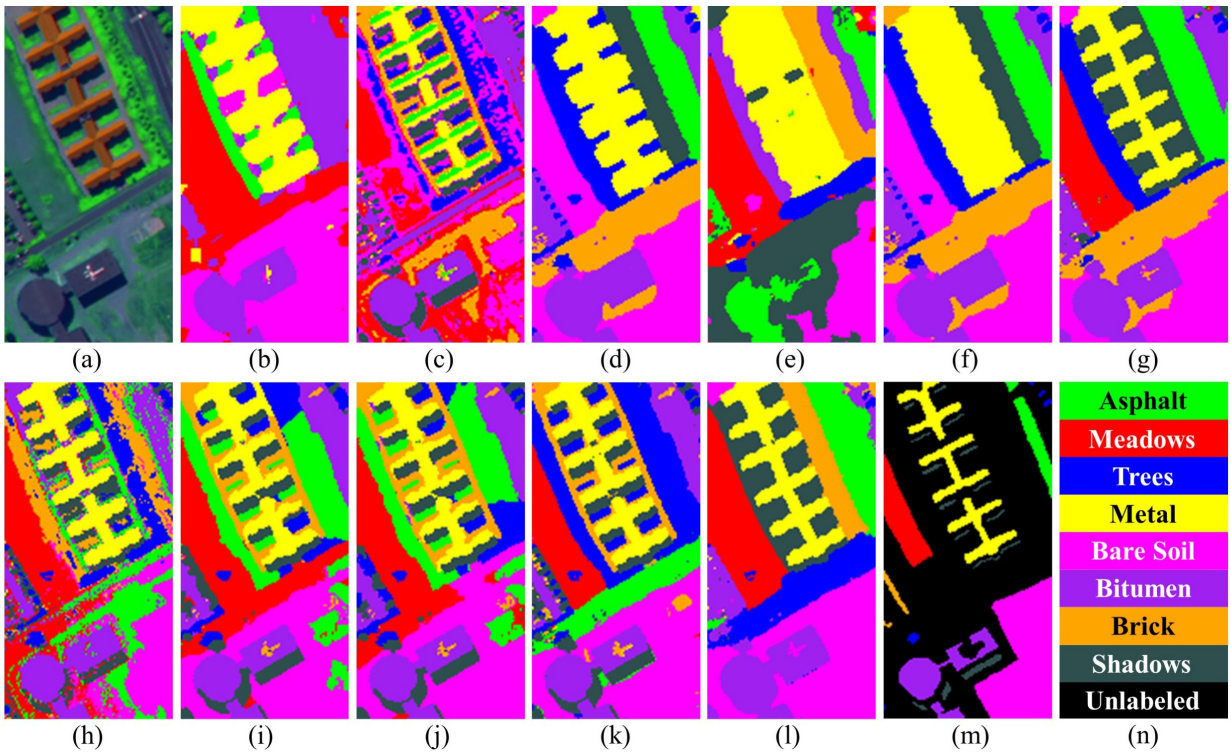


Fig. 6. Classification maps of the different methods with the PUS dataset (with background): (a) false color image, (b) EKGCSC, (c) SSC, (d) DSC, (e) DS^{3}C, (f) S^{2}ConvSCN, (g) SDSC-AI, (h) JSSC, (i) SSC+, (j) FHoSSC, (k) DS-GCSC, (l) our proposed DLR-GCSC, (m) ground truth, and (n) label.

visual comparison between different classification methods, as shown in Figs. 5–7. It can be seen that our proposed DLR-GCSC achieves the best classification results, which outperforms both the traditional subspace clustering methods and DSC methods on three datasets. The classification result of SSC is relatively poor in that it belongs to a pixel-level method. By introducing superpixel segmentation and spectral–spatial information, JSSC, SSC+, and FHoSSC achieve better classification performance. EKGCSC and other existing DSC methods perform better than SSC but worse than JSSC, SSC+, and FHoSSC. This is mainly because these methods are designed for HSI datasets without background to guarantee that the latent features are relatively pure and robust. However, when faced with the HSI datasets with background, the existing DSC methods are even difficult to reach the performance of the traditional SSC-based methods. By utilizing the DLR joint feature extraction and GCSC, our proposed method could solve this problem and further improve the classification performance. On the IPS dataset, our proposed DLR-GCSC achieves the best classification result with the OA of 92.19%, the AA of 92.52%, and the Kappa of 88.90%. On the PUS dataset, DLR-GCSC achieves the best classification result with the OA of 95.95%, the AA of 79.95%, and the Kappa of 94.56%. On the SA dataset, DLR-GCSC also achieves the best classification result with the OA of 99.70%, the AA of 99.62%, and the Kappa of 99.63%. Due to the fact that there are very small numbers of Bricks category in the PUS dataset, it is hard to classify the Bricks category, which results in a relatively low AA of all the compared methods on the PUS dataset (all of these methods are lower than 80%).

C. Results and Analysis on HSI Datasets Without Background

To demonstrate the generalization capability of our proposed method, we conduct another experiment on the IPS, PUS,

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TABLE V

CLASSIFICATION RESULTS OF EKGCSC, SSC, DSC, DS^{3}C, S^{2}CONVSCN, SDSC-AI, JSSC, SSC+, FHOSSC, DGCSC, AND DLR-GCSC ON THE SA DATASET

Class	EKGCSC	SSC	DSC	DS^{3}C	S^{2}ConvSCN	SDSC-AI	JSSC	SSC+	FHoSSC	DGCSC	DLR-GCSC
Brocoli_green_weeds_1(%)	90.79	14.07	99.49	100	100	99.74	99.49	100	100	99.74	99.49
Corn_senesced_green(%)	77.14	56.96	43.04	57.19	60.83	67.39	99.26	95.23	95.31	80.34	100
Lettuce_romaine_4wk(%)	85.71	92.86	100	99.35	99.84	95.78	94.97	96.92	96.75	100	100
Lettuce_romaine_5wk(%)	88.72	99.41	100	100	100	100	100	99.93	100	99.93	100
Lettuce_romaine_6wk(%)	100	100	99.70	98.37	100	100	99.85	100	100	99.85	99.85
Lettuce_romaine_7wk(%)	0.00	95.87	98.25	100	99.50	100	97.37	100	100	99.75	98.75
$OA(\%)$	73.78	81.30	85.36	88.97	90.07	91.31	98.78	98.43	98.45	94.97	**99.70**
$AA(\%)$	73.73	76.53	90.08	92.49	93.36	93.82	98.49	98.68	98.68	96.60	**99.62**
$Kappa(\%)$	67.25	76.98	82.10	86.37	87.72	89.22	98.48	98.04	98.06	93.74	**99.63**

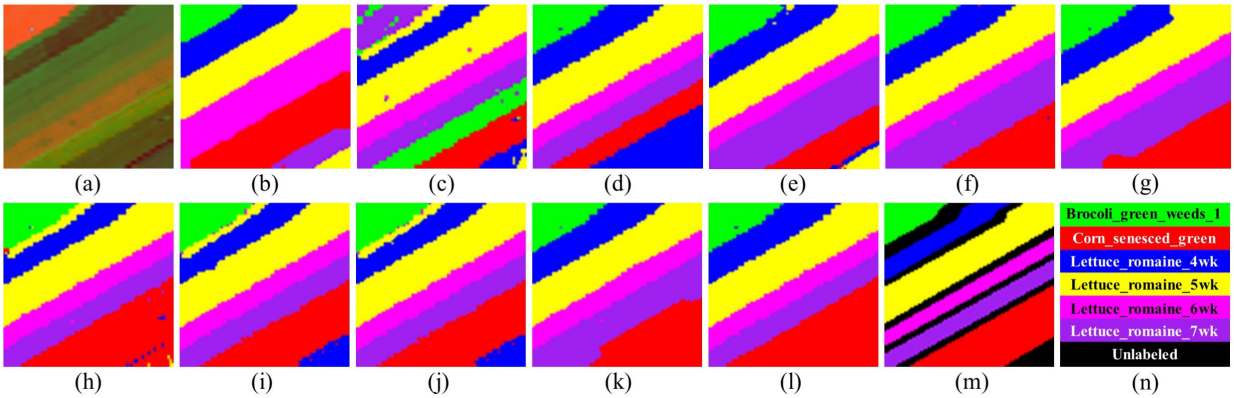


Fig. 7. Classification maps of the different methods with the SA dataset (with-background): (a) false color image, (b) EKGCSC, (c) SSC, (d) DSC, (e) DS^{3}C, (f) S^{2}ConvSCN, (g) SDSC-AI, (h) JSSC, (i) SSC+, (j) FHoSSC, (k) DS-GCSC, (l) our proposed DLR-GCSC, (m) ground truth, and (n) label.

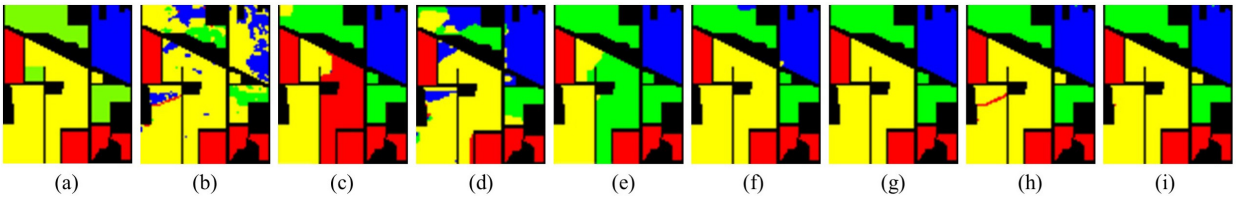


Fig. 8. Classification maps of the different methods on with the IPS-nbk dataset: (a) ground truth, (b) RMMF, (c) DSC, (d) DS^{3}C, (e) S^{2}ConvSCN, (f) EKGCSC, (g) SDSC-AI, (h) DGCSC, and (i) our proposed DLR-GCSC.

and SA datasets without background (we call them IPS-nbk, PUS-nbk, and SA-nbk below). The quantitative classification results, as shown in Table VI, and the visual classification maps can be seen in Figs. 8–10. It can be clearly observed that the classification results of existing DSC-based methods (DSC, DS^{3}C, S^{2}ConvSCN, SDSC-AI, DGCSC, and DLR-GCSC) and traditional subspace clustering method EKGCSC on HSI datasets without background are much better than their performance on HSI datasets with background, which demonstrates that the background information, as a type of noise information, can exactly cause the degradation of classification performance. In addition, our proposed method has reached the top classification results (OA of 100%, AA of 100%, and Kappa of 100%) on PUS-nbk and SA-nbk datasets. Although SDSC-AI has also reached 100% classification result on PUS-nbk and SA-nbk datasets, our proposed method DLR-GCSC improves the OA, AA, and Kappa by 0.7%, 0.31%, and 1.98% compared with SDSC-AI, respectively, on the IPS-nbk dataset and achieves the best classification result.

D. Parameters Analysis

1) Impact of the Regularization Parameters λ and γ : In this experiment, we explore the effect of the tradeoff parameters λ and γ on the classification performance of DLR-GCSC with the other parameters fixed; the rest two parameters α and β are both set to 0.5 as previously mentioned. Given $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}\}$ for IPS and PUS datasets, $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}\}$ for the SA dataset, and $\gamma \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}\}$ for all three datasets, the relationship between OA and two parameters is shown in Fig. 12. When $\lambda = 10^{0}$ and $\gamma = 10^{2}$, DLR-GCSC can achieve the highest OA on the IPS dataset. Similarly, $\lambda = 10^{-3}$ and $\gamma = 10^{2}$ are the best choice for the PUS dataset, and $\lambda = 10^{-4}$ and $\gamma = 10^{2}$ are selected for the SA dataset to get the best OA result.

2) Impact of the Nearest Neighbors k: The nearest neighbors k is used to construct the graph embedding for the HSI dataset. In this experiment, we investigate the influence of k with the other parameters fixed. Fig. 11(a) shows the change in OA corresponding to different PCs for the three datasets.

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TABLE VI

CLASSIFICATION RESULTS OF RMMF, DSC, DS^{3}C, S^{2}CONVSCN, EKGCSC, SDSC-AI, DGCSC, AND DLR-GCSC ON THREE HSI DATASETS WITHOUT BACKGROUND

Dataset	Metric	RMMF	DSC	DS^{3}C	S^{2}ConvSCN	EKGCSC	SDSC-AI	DGCSC	DLR-GCSC
IPS-nbk	$OA(\%)$	72.21	76.66	83.88	77.09	96.72	97.43	97.59	**98.13**
	$AA(\%)$	66.04	85.26	84.58	86.28	96.76	97.58	97.65	**97.96**
	$Kappa(\%)$	56.09	68.88	79.89	69.16	89.25	91.86	92.31	**94.29**
PUS-nbk	$OA(\%)$	77.04	93.48	86.87	98.54	97.35	**100**	**100**	**100**
	$AA(\%)$	48.36	85.71	56.83	88.67	84.69	**100**	**100**	**100**
	$Kappa(\%)$	68.04	91.61	86.85	98.08	95.24	**100**	**100**	**100**
SA-nbk	$OA(\%)$	98.20	88.44	99.79	99.23	**100**	**100**	**100**	**100**
	$AA(\%)$	98.17	83.58	99.65	99.49	**100**	**100**	**100**	**100**
	$Kappa(\%)$	97.75	85.27	99.20	99.04	**100**	**100**	**100**	**100**

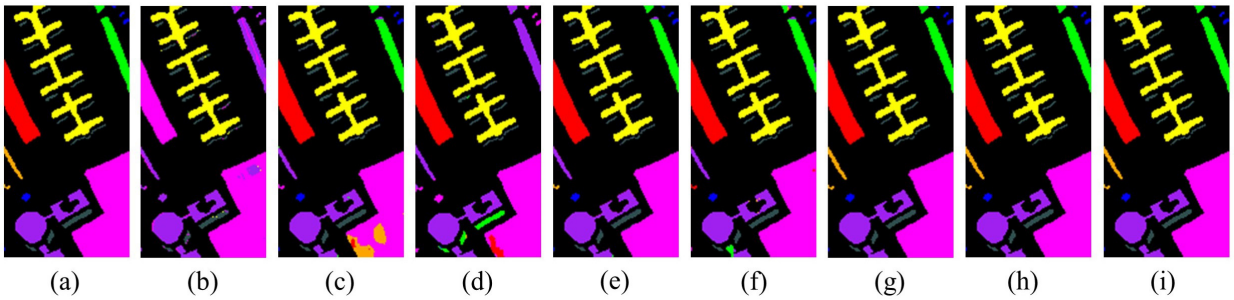


Fig. 9. Classification maps of the different methods on with the PUS-nbk dataset: (a) ground truth, (b) RMMF, (c) DSC, (d) DS^{3}C, (e) S^{2}ConvSCN, (f) EKGCSC, (g) SDSC-AI, (h) DGCSC, and (i) our proposed DLR-GCSC.

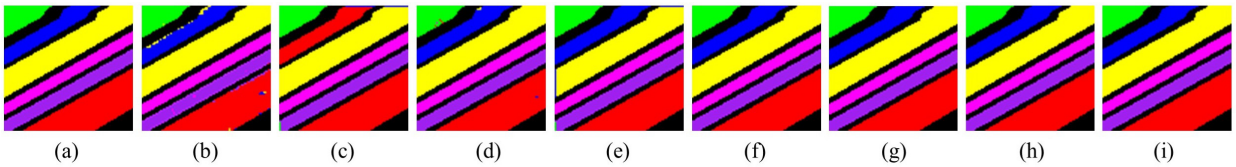


Fig. 10. Classification maps of the different methods on with the SA-nbk dataset: (a) ground truth, (b) RMMF, (c) DSC, (d) DS^{3}C, (e) S^{2}ConvSCN, (f) EKGCSC, (g) SDSC-AI, (h) DGCSC, and (i) our proposed DLR-GCSC.

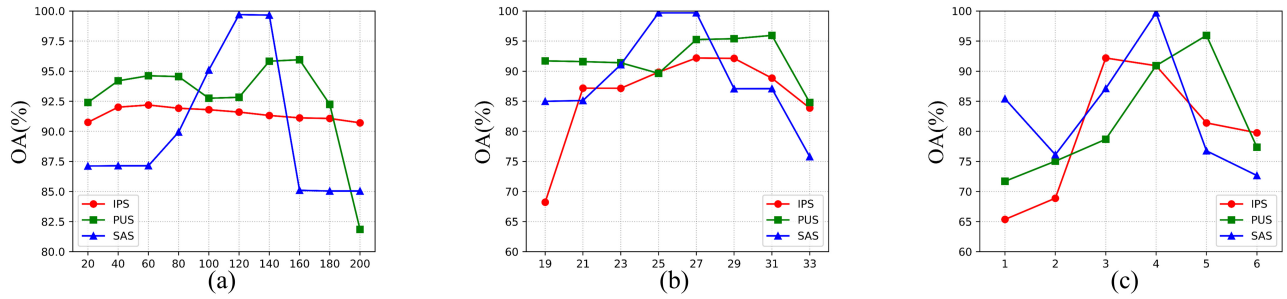


Fig. 11. OA comparison under different parameters for our proposed DLR-GCSC on IPS, PUS, and SA datasets: (a) nearest neighbors k, (b) patch window size w, and (c) principal components d.

It can be obviously seen that the OA rises first to the top and then falls with the k increasing. This is mainly because, if k becomes too large, the oversmoothing problem will happen and, finally, lead to the decrease of OA. Therefore, we set k to 60, 120, and 160 for IPS, PUS, and SA datasets, respectively.

3) Impact of the Patch Window Size w: The patch window size w is the size of the spatial patches that are fed into the 2-D CAE to extract the spatial information. We conduct the experiment with the other parameters fixed to explore the impact of w. As shown in Fig. 11(b), if w is set too small, the 2-D CAE structure may be not likely to extract enough spatial information of HSI. On the contrary, the patches could contain more redundant spatial information, which is a disadvantage for improving the classification result. The best performance is achieved at $w = 27, 31, 27$ for IPS, PUS, and SA datasets, respectively.

4) Impact of the Number of Principal Components d: In this experiment, we evaluate the effect of the number of principal components with the other parameters fixed. The number of PCs is equal to the dimension reduced by

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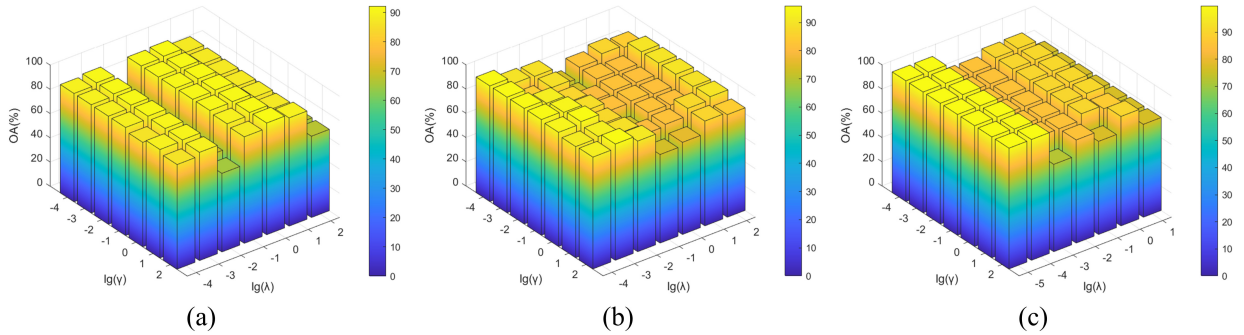


Fig. 12. OA comparison of λ and γ for our proposed DLR-GCSC on (a) IPS, (b) PUS, and (c) SA datasets.

TABLE VII

CLASSIFICATION RESULTS OF DLR-GCSC_1D, DLR-GCSC_2D, DLR-SC, AND DLR-GCSC

Dataset	IPS			PUS			SA		
	$OA(\%)$	$AA(\%)$	$Kappa(\%)$	$OA(\%)$	$AA(\%)$	$Kappa(\%)$	$OA(\%)$	$AA(\%)$	$Kappa(\%)$
DLR-GCSC_1D	71.51	63.64	55.76	62.93	58.37	53.71	78.85	73.32	79.51
DLR-GCSC_2D	80.80	84.35	73.42	81.19	77.46	76.23	85.55	81.44	88.66
DLR-SC	82.78	83.59	75.62	81.58	68,16	76.82	88.39	83.57	88.77
DLR-GCSC	**92.19**	**92.52**	**88.90**	**95.95**	**79.95**	**94.65**	**99.70**	**99.62**	**99.63**

TABLE VIII

RUNNING TIME COMPARISON OF EKGCSC, SSC, DSC, DS^{3}C, S^{2}CONVSCN, SDSC-AI, JSSC, SSC+, FHOSSC, DGCSC, AND DLR-GCSC ON IPS, PUS, AND SA DATASETS (IN SECOND)

Dataset	EKGCSC	SSC	JSSC	SSC+	FHoSSC	DSC	DS^{3}C	S^{2}ConvSCN	SDSC-AI	DGCSC	DLR-GCSC
IPS(s)	126.23	624.61	57.33	70.25	20.12	700.47	742.11	813.69	765.87	970.66	897.63
PUS(s)	436.72	2037.45	277.46	306.55	124.20	2456.93	2567.47	2706.16	2783.22	3217.31	3016.87
SA(s)	144.59	834.23	89.54	101.89	45.39	917.64	965.24	1022.70	988.53	1149.36	1054.35

PCA; we perform our DLR-GCSC with a range of PCs from 1 to 6 for three datasets. The corresponding OA is shown in Fig. 11(c). When the number of PCs is set to 3, 4, and 5, the IPS, PUS, and SA datasets reach their best classification result, respectively. Note that, as the number of PCs increases too large, the classification result begins to decrease because of the phenomenon called the Hughes phenomenon [50].

E. Ablation Study

To demonstrate the effectiveness of our proposed method, we conduct the ablation study on three datasets. DLR-GCSC only with 1-D band CAE (DLR-GCSC_1D), DLR-GCSC only with 2-D patch CAE (DLR-GCSC_2D), and DLR-SC are used as the comparison methods. Table VII shows the classification results of DLR-GCSC_1D, DLR-GCSC_2D, and DLR-SC on IPS, PUS, and SA datasets. It could be observed that, compared with the single CAE structure (DLR-GCSC_1D and DLR-GCSC_2D), the joint spatial–spectral CAE structure (DLR-GCSC) has a significant improvement in terms of OA, AA, and Kappa on three datasets. Compared with DLR-SC, the classification results of DLR-GCSC also improve a lot because DLR-GCSC has considered the latent graph-based global information. Fig. 13 shows the visualization of the affinity matrix on the SA dataset. We can observe that the affinity matrix obtained from DLR-GCSC has a clearer block-diagonal structure than obtained from DLR-SC, which demonstrates

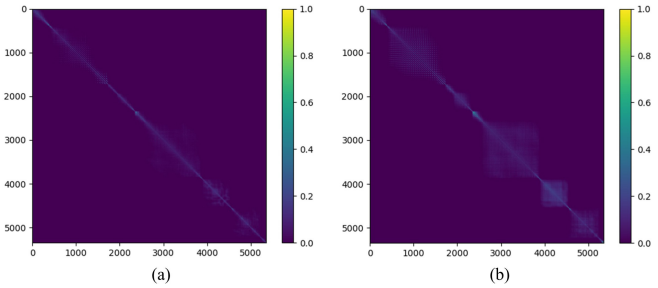


Fig. 13. Visualization of the obtained affinity matrix on the SA dataset: (a) DLR-SC and (b) DLR-GCSC.

that DLR-GCSC can describe accurate intrinsic relationships between all data points.

F. Comparison of Running Time

The running time of different compared methods on three HSI datasets with background is shown in Table VIII. The superpixel-level methods (e.g., JSSC, SSC+, and FHoSSC) have a relatively lower time consumption than other compared methods because the superpixel-level methods group pixels into perceptually meaningfully atomic regions, which equals to alleviate the computational cost by reducing the number of inputs. Among the pixel-level methods (e.g., EKGCSC, SSC, DSC, DS^{3}C, S^{2}ConvSCN, and SDSC-AI), SSC has a low classification accuracy and high time-consuming because it is the most traditional method without any feature extraction. DSC,

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DS^{3}C, S^{2}ConvSCN, and SDSC-AI extract the spatial–spectral features by using 2-D CAE, and the self-supervise technic is adopted to further enhance the training of feature learning. They bring the burden to the running time and have limited OA improvement on the HSI dataset with background. The running time of the proposed DLR-GCSC is a little slower than other compared methods, but the classification performance has been improved greatly in that our joint spatial–spectral network can learn discriminative joint features. Compared with DGCSC that uses the original DSC self-expressive layer, DLR-GCSC has lower running time and higher classification accuracy, which demonstrates the effectiveness of the DLR layer.

V. Conclusion

In this article, we propose an end-to-end DLR-GCSC framework for HSI classification, in which the HSI dataset with background is available for model training. Compared with other existing DSC methods, we add a 1-D CAE and combine it with the original 2-D CAE to construct a joint spatial–spectral network, which can make full use of the spatial–spectral information. To make the joint features more discriminative, a low-rank constrained fully connected layer is designed as the self-expressive layer in the joint spatial–spectral network. By introducing graph convolution, the latent graph-based global information of HSI is considered to further improve the classification performance. At the same time, the GCSC alleviates the performance degradation caused by noisy samples (background). Experiments on three benchmark HSI datasets demonstrated that our method can obtain the best classification results *not only* on the datasets with background *but also* on the datasets without background.

In the learning of the affinity matrix, the large-scale HSI datasets could cause a high computational and memory consumption because the construction of the corresponding affinity matrix needs all samples totally. In future work, we try to explore a collaborative learning strategy to address the high-cost problem on large-scale HSI datasets.

References

[1] A. Plaza *et al.*, "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Sep. 2009.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] B. Datt, T. R. McVicar, T. G. Van Niel, D. L. B. Jupp, and J. S. Pearlman, "Preprocessing EO-1 Hyperion hyperspectral data to support the application of agricultural indexes," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1246–1259, Jun. 2003.

[4] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.

[5] T. Matsuki, N. Yokoya, and A. Iwasaki, "Hyperspectral tree species classification of Japanese complex mixed forest with the aid of lidar data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 2177–2187, May 2015.

[6] C. Ke, "Military object detection using multiple information extracted from hyperspectral imagery," in *Proc. Int. Conf. Prog. Informat. Comput. (PIC)*, Dec. 2017, pp. 124–128.

[7] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.

[8] P. Jia, M. Zhang, W. Yu, F. Shen, and Y. Shen, "Convolutional neural network based classification for hyperspectral data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5075–5078.

[9] Y. Zhan, D. Hu, Y. Wang, and X. Yu, "Semisupervised hyperspectral image classification based on generative adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 212–216, Feb. 2018.

[10] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[11] J. Feng, H. Yu, L. Wang, X. Cao, X. Zhang, and L. Jiao, "Classification of hyperspectral images based on multiclass spatial-spectral generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5329–5343, Mar. 2019.

[12] Z. Zhong, J. Li, D. A. Clausi, and A. Wong, "Generative adversarial networks and conditional random fields for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 50, no. 7, pp. 3318–3329, Jul. 2020.

[13] H. Sun, X. Zheng, and X. Lu, "A supervised segmentation network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 2810–2825, 2021.

[14] Z. Gong, P. Zhong, and W. Hu, "Statistical loss and analysis for deep learning in hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 322–333, Jan. 2021.

[15] Z. Gong, W. Hu, X. Du, P. Zhong, and P. Hu, "Deep manifold embedding for hyperspectral image classification," *IEEE Trans. Cybern.*, early access, Apr. 19, 2021, doi: 10.1109/TCYB.2021.3069790.

[16] S. Lloyd, "Least squares quantization in PCM," *IEEE Trans. Inf. Theory*, vol. IT-28, no. 2, pp. 129–137, Mar. 1982.

[17] J. C. Bezdek, *Pattern Recognition With Fuzzy Objective Function Algorithms*. Cham, Switzerland: Springer, 2013.

[18] J. J. Buckley, "Fuzzy hierarchical analysis," *Fuzzy Sets Syst.*, vol. 17, no. 3, pp. 233–247, Dec. 1985.

[19] L. Zhang, L. Zhang, B. Du, J. You, and D. Tao, "Hyperspectral image unsupervised classification by robust manifold matrix factorization," *Inf. Sci.*, vol. 485, pp. 154–169, Jun. 2019.

[20] R. Vidal, "Subspace clustering," *IEEE Signal Process. Mag.*, vol. 28, no. 2, pp. 52–68, Mar. 2011.

[21] L. Lu and R. Vidal, "Combined central and subspace clustering for computer vision applications," in *Proc. 23rd Int. Conf. Mach. Learn.*, 2006, pp. 593–600.

[22] J. Chen and J. Yang, "Robust subspace segmentation via low-rank representation," *IEEE Trans. Cybern.*, vol. 44, no. 8, pp. 1432–1445, Aug. 2014.

[23] G. Liu, Z. Lin, S. Yan, J. Sun, Y. Yu, and Y. Ma, "Robust recovery of subspace structures by low-rank representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 171–184, Jan. 2013.

[24] R. Vidal and P. Favaro, "Low rank subspace clustering (LRSC)," *Pattern Recognit. Lett.*, vol. 43, pp. 47–61, Jul. 2014.

[25] C.-Y. Lu, H. Min, Z.-Q. Zhao, L. Zhu, D.-S. Huang, and S. Yan, "Robust and efficient subspace segmentation via least squares regression," in *Proc. Eur. Conf. Comput. Vis.*, 2012, pp. 347–360.

[26] P. Ji, M. Salzmann, and H. Li, "Efficient dense subspace clustering," in *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, Mar. 2014, pp. 461–468.

[27] L. Tian, Q. Du, and I. Kopriva, "ℓ_0-motivated low rank sparse subspace clustering for hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 1038–1041.

[28] C.-G. Li, C. You, and R. Vidal, "Structured sparse subspace clustering: A joint affinity learning and subspace clustering framework," *IEEE Trans. Image Process.*, vol. 26, no. 6, pp. 2988–3001, Jun. 2017.

[29] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[30] S. Huang, H. Zhang, and A. Pizurica, "Joint sparsity based sparse subspace clustering for hyperspectral images," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3878–3882.

[31] L. Wang *et al.*, "Fast high-order sparse subspace clustering with cumulative MRF for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 152–156, Jan. 2021.

[32] J. Bruna, W. Zaremba, A. Szlam, and Y. LeCun, "Spectral networks and locally connected networks on graphs," in *Proc. Int. Conf. Learn. Represent.*, 2014, pp. 1–14.

[33] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3844–3852.

[34] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, 2017, pp. 1–14.

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[35] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2020.
[36] P. Ji, T. Zhang, H. Li, M. Salzmann, and I. Reid, "Deep subspace clustering networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30. Red Hook, NY, USA: Curran Associates, 2017.
[37] Y. Long, X. Deng, G. Zhong, J. Fan, and F. Liu, "Gaussian kernel dynamic similarity matrix based sparse subspace clustering for hyperspectral images," in *Proc. 15th Int. Conf. Comput. Intell. Secur. (CIS)*, Dec. 2019, pp. 211–215.
[38] S. Xiao, M. Tan, D. Xu, and Z. Y. Dong, "Robust kernel low-rank representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 11, pp. 2268–2281, Nov. 2016.
[39] P. Zhou, Y. Hou, and J. Feng, "Deep adversarial subspace clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1596–1604.
[40] J. Zhang *et al.*, "Self-supervised convolutional subspace clustering network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5468–5477.
[41] M. Kheirandishfard, F. Zohrizadeh, and F. Kamangar, "Deep low-rank subspace clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 3776–3781.
[42] M. Zeng, Y. Cai, X. Liu, Z. Cai, and X. Li, "Spectral-spatial clustering of hyperspectral image based on Laplacian regularized deep subspace clustering," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2694–2697.
[43] J. Lei, X. Li, B. Peng, L. Fang, N. Ling, and Q. Huang, "Deep spatial-spectral subspace clustering for hyperspectral image," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 7, pp. 2686–2697, Jul. 2021.
[44] K. Li, Y. Qin, Q. Ling, Y. Wang, Z. Lin, and W. An, "Self-supervised deep subspace clustering for hyperspectral images with adaptive self-expressive coefficient matrix initialization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3215–3227, 2021.
[45] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.
[46] E. Elhamifar and R. Vidal, "Sparse subspace clustering: Algorithm, theory, and applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2765–2781, Nov. 2013.
[47] S. Boyd, N. Parikh, and E. Chu, *Distributed Optimization and Statistical Learning Via the Alternating Direction Method of Multipliers*. Boston, MA, USA: Now, 2011.
[48] J. Munkres, "Algorithms for the assignment and transportation problems," *J. Soc. Ind. Appl. Math.*, vol. 5, no. 1, pp. 32–38, 1957.
[49] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–13.
[50] G. V. Trunk, "A problem of dimensionality: A simple example," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. PAMI-1, no. 3, pp. 306–307, Jul. 1979.



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